

Producer Theory

Lecture 13: The Seller's Supply Rule — $\text{Price} = \text{Marginal Cost}$

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Recap: Lecture 12

What we covered last time:

- **Shutdown rule:** Operate if $P \geq AVC$; shut down if $P < AVC$
- **Breakeven price** = $ATC_{\min}$ (zero economic profit)
- **Shutdown price** = $AVC_{\min}$ (below this, stop producing)
- **The firm's supply curve** = the MC curve above AVC

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 **Today:** We formalize the **seller's supply rule** and explore what makes the supply curve **shift**.

 The supply rule is the **producer's equivalent** of the demand curve we derived in Lecture 8!

The Seller's Supply Rule

The Cost-Benefit Principle Applied to Firms

THE SELLER'S SUPPLY RULE

A profit-maximizing firm in perfect competition should produce at the level of output where:

$$\text{Price} = \text{Marginal Cost} \quad (P = MC)$$

on the **rising** portion of the MC curve, provided $P \geq AVC$.

Why? The **cost-benefit principle** (from Lecture 3!) applied to each unit:

- The **benefit** of producing one more unit = the revenue it brings = P
- The **cost** of producing one more unit = MC

👉 Keep producing as long as $P \geq MC$. Stop when the next unit would cost more than it earns.

This is just **marginal analysis** — the same logic we've used since the very beginning of the course!

The Rule in Action: Step by Step

The bottle factory (from the textbook): $P = €0.35/\text{bottle}$, $FC = €40/\text{day}$

Going from → to	Extra bottles	Revenue per bottle (MR = P)	MC per bottle	MR vs MC	Decision
0 → 100	100	€0.35	€0.10	MR > MC	✓ Produce
100 → 200	100	€0.35	€0.10	MR > MC	✓ Produce
200 → 300	100	€0.35	€0.20	MR > MC	✓ Produce
300 → 400	100	€0.35	€0.30	MR > MC	✓ Produce
400 → 500	100	€0.35	€0.40	MR < MC	✗ Stop!

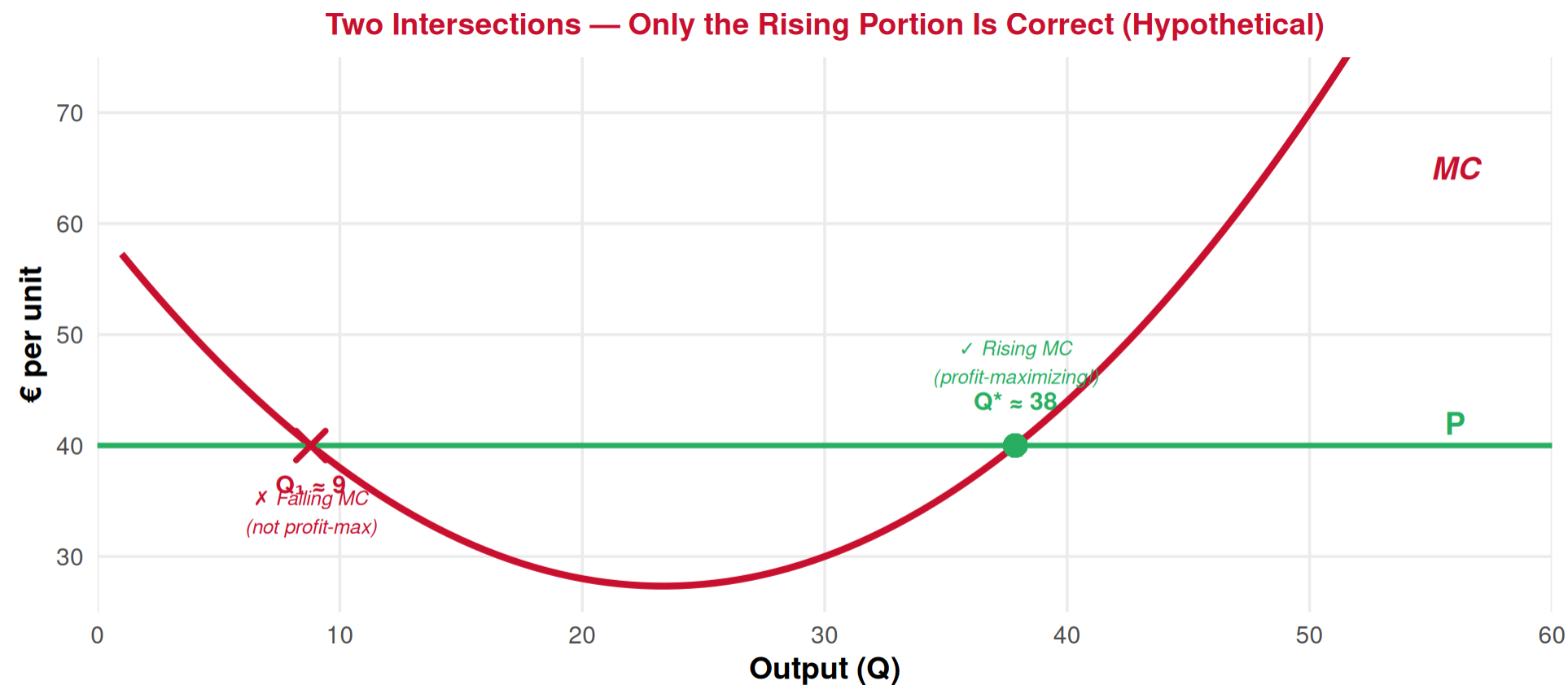
Source: Course textbook (sebenta), Table 10

👉 **Optimal output: $Q^* = 400$ bottles/day**. At $Q = 400$, the MC of the next batch (€0.40) exceeds the price (€0.35), so expanding further would **reduce** profit.

💡 At $Q = 300$, $MC = €0.20 < P = €0.35$ — there's still €0.15/bottle of “profit margin” to capture. That's why the firm doesn't stop at 300!

Why “Rising Portion” Matters

MC is U-shaped — it may intersect the price line **twice**. Which intersection is correct?



At Q_1 (falling MC): if you produce one more unit, MC is still falling — you can **increase** profit further! At Q^* (rising MC): producing one more unit would cost **more** than the price. This is the true optimum.

The Supply Rule: Formal Summary

THE COMPLETE SUPPLY RULE FOR A COMPETITIVE FIRM

1. Find Q^* where $P = MC$ on the **rising** portion of MC
2. If $P \geq AVC$ at $Q^* \rightarrow$ **supply Q^* units**
3. If $P < AVC$ at $Q^* \rightarrow$ **supply 0 units** (shut down)

The individual supply curve is the **MC curve above AVC_{min}** .

This rule answers three questions at once:

Question	Answer
How much to produce?	Where $P = MC$ (rising portion)
Whether to produce?	Only if $P \geq AVC$
How much profit?	$\pi = (P - ATC) \times Q^*$

 One rule, three answers — that's the power of marginal analysis!

What Happens When Costs Change?

The Textbook's Key Insight: Changing the Price

The textbook shows what happens when the bottle price **risers from €0.35 to €0.45**:

Q (bottles/day)	MC per bottle	Profit at P=€0.35	Profit at P=€0.45
0	—	−40	−40
100	0.10	−15	−5
200	0.10	10	30
300	0.20	25	55
400	0.30	30	70
500	0.40	25	75
600	0.50	10	70
700	0.60	−15	55

Source: Course textbook (sebenta), Tables 10 & 11

- At P = €0.35: **$Q^* = 400$** (MC ≈ 0.30 , next batch MC = 0.40 > P)
- At P = €0.45: **$Q^* = 500$** (MC ≈ 0.40 , next batch MC = 0.50 > P)

 **Higher price $\rightarrow$ produce more $\rightarrow$ earn more profit.** The firm moves **up along its MC/supply curve**.

The Textbook's Key Insight: Changing Wages

What if wages rise from €10/hour to **€12/hour** (keeping $P = €0.35$)?

Q	LC at €10/hr	LC at €12/hr	TC (w=10)	TC (w=12)	Profit (w=10)	Profit (w=12)
0	0	0	40	40	-40	-40
100	10	12	50	52	-15	-17
200	20	24	60	64	10	6
300	40	48	80	88	25	17
400	70	84	110	124	30	16
500	110	132	150	172	25	3

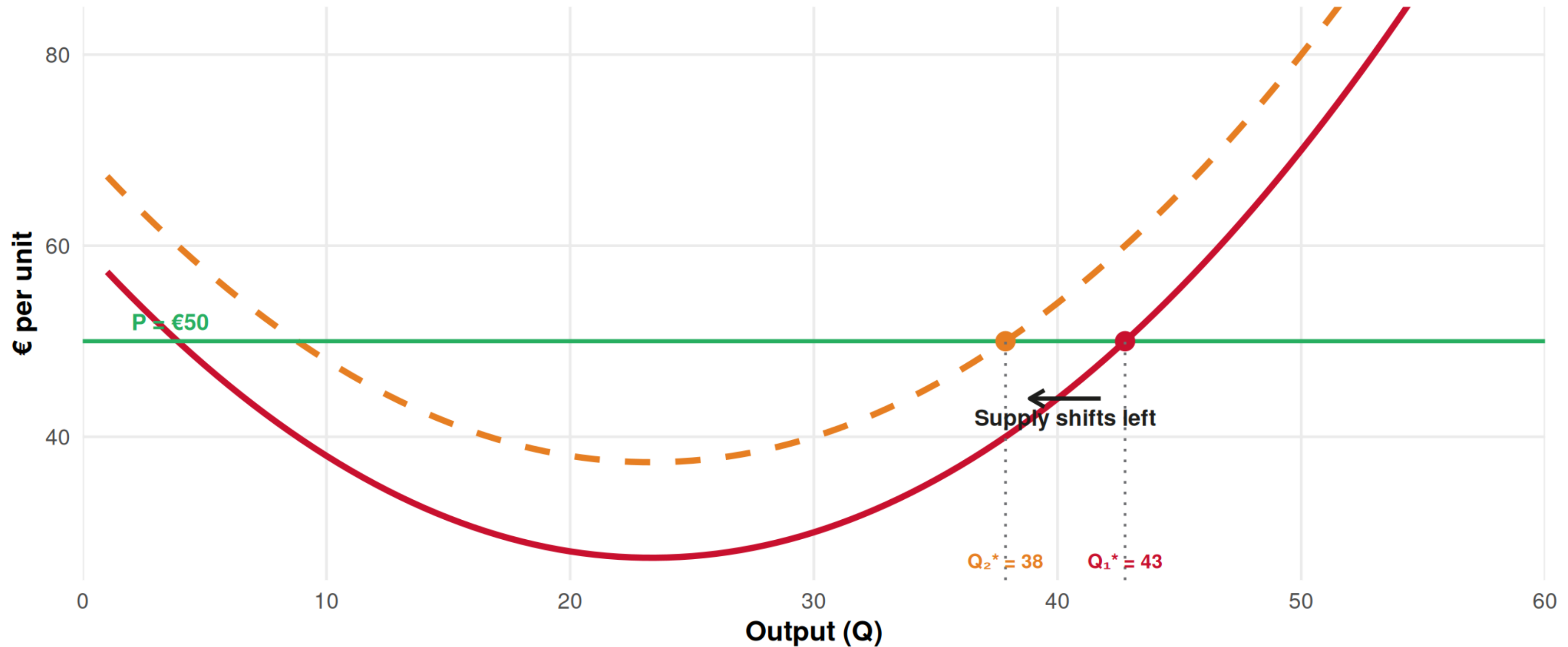
Source: Course textbook (sebenta), Tables 10 & 12

- At $w = €10/\text{hr}$: **$Q^* = 400$** , max profit = €30
- At $w = €12/\text{hr}$: **$Q^* = 300$** , max profit = €17

 **Higher wages $\rightarrow$ MC rises at every Q $\rightarrow$ firm produces less at the same price.** The supply curve has **shifted left!**

Visualizing a Supply Shift

Effect of Higher Wages: Supply Shifts Left (Hypothetical)



Does Changing Fixed Costs Shift Supply?

The textbook also shows: when **FC rises from €40 to €70** (price stays at €0.35):

- **Q^* stays at 400!** The profit-maximizing quantity doesn't change.
- Profit drops from €30 to €0, but the **production decision** is unchanged.

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FIXED COSTS DO NOT SHIFT THE SUPPLY CURVE IN THE SHORT RUN

MC depends only on **variable costs**. Since FC doesn't affect MC, it doesn't change the $P = MC$ intersection, and the supply curve stays in the same position.

FC affects **profit** (and the shutdown/exit decision in the long run), but not the **quantity supplied** at each price.

👉 **Movement along** the supply curve: caused by a change in the product's **own price**

👉 **Shift of** the supply curve: caused by a change in **input prices** (wages, materials), **technology**, or **number of firms**

Supply Curve Shifters

What Shifts the Supply Curve?



Shift RIGHT (increase in supply) →

More is supplied at every price when:

-  **Input prices fall** (cheaper labor, fuel, materials)
-  **Technology improves** (online booking systems, automation)
-  **Favorable conditions** (good weather for agriculture, tourism)
-  **More firms enter** the market
-  **Government subsidies** or reduced regulation

Tourism: New low-cost airline tech → cheaper flights → supply of flights shifts right

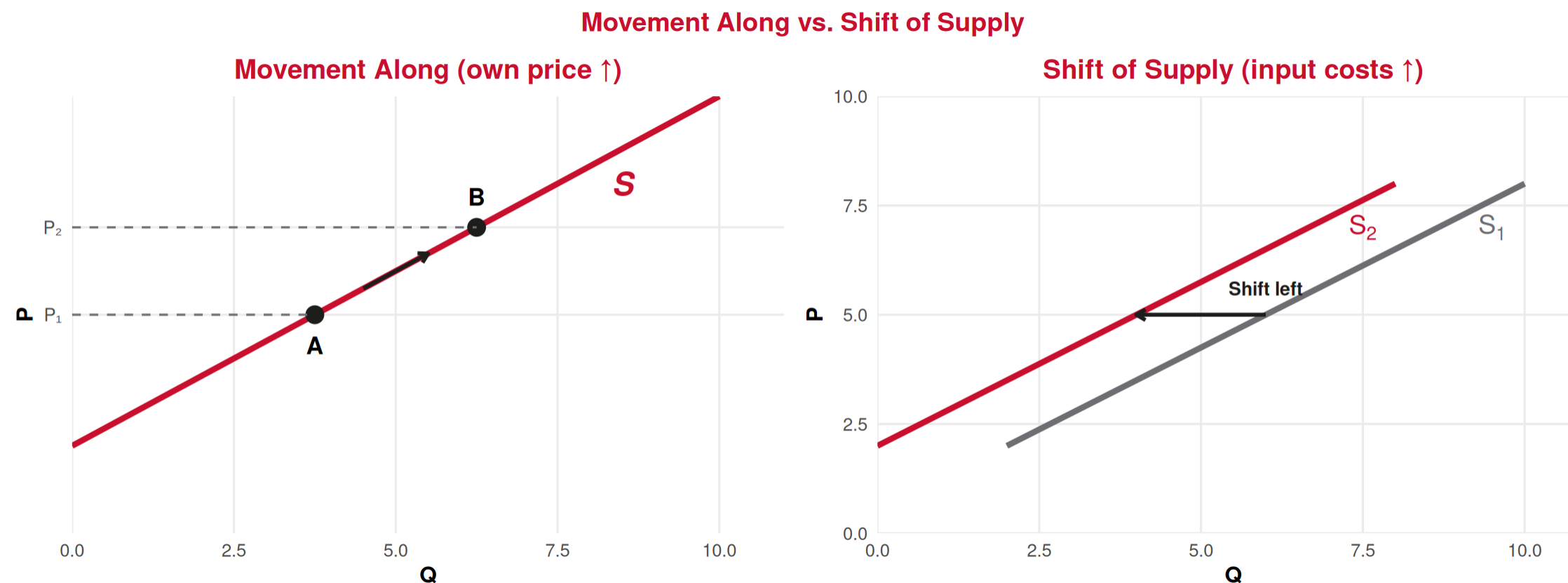
Shift LEFT (decrease in supply) ←

Less is supplied at every price when:

-  **Input prices rise** (higher wages, energy costs)
-  **Regulation increases** (stricter licensing, environmental rules)
-  **Adverse events** (natural disaster, pandemic)
-  **Firms exit** the market
-  **Higher taxes** on producers

Tourism: New tourist tax → higher costs for hotels → supply shifts left

Movement Along vs Shift: Supply Edition



Left: When the product's own price rises, the firm moves **along** its supply curve → Q increases.

Right: When input costs rise (wages, energy), the entire supply curve **shifts left** → less supplied at every price.



Exactly the same logic as demand (Lecture 8): own-price = movement along; other factors = shift!

Tourism Applications

Supply Shifts in Portuguese Tourism

Example 1: Minimum wage increase

Portugal's minimum wage has been rising. For tourism businesses:

- Labor is a major variable cost (housekeeping, waitstaff, guides)
- Higher wages → MC shifts up → supply shifts **left**
- At the same hotel room price, fewer rooms are profitably offered
- Some marginal businesses may shut down ($P < AVC$ at new costs)

Effect: fewer beds at every price level, or higher prices to compensate

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 Both examples change **variable costs**, which shifts MC and therefore the supply curve.

Example 2: Online booking technology

Platforms like Booking.com, Airbnb:

- Reduce distribution costs (no need for travel agents)
- Lower marketing costs per booking
- MC shifts **down** → supply shifts **right**
- More accommodation available at every price point

Effect: more rooms at every price level — which has been the trend!

Supply Shifts: Seasonal Staffing in the Algarve

Why is the supply of tourism services different in summer vs winter?

The $P = MC$ rule explains it:

 **Summer:**

- Abundant seasonal workers → wages moderate
- MC is **relatively low** for extra services
- Supply curve is **further right**
- More tours, restaurants, activities offered

 **Winter:**

- Workers move to other sectors → must pay more to retain
- Energy costs higher (heating)
- MC is **higher** for the same services
- Supply curve shifts **left**
- Fewer services offered

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 **Combined with demand shifts:** In summer, both supply and demand are higher. In winter, both shift left. The net effect on price depends on which shift is larger — this is **market equilibrium** (Lecture 17)!

The Parallel with Demand

Throughout the course, we've built **mirror-image** tools for consumers and producers:

point_right: Next lecture we complete the right column: **producer surplus** and **market supply**!

Concept	Consumer (Demand)	Producer (Supply)
Goal	Maximize utility	Maximize profit
Constraint	Budget (M)	Cost structure (TC)
Individual curve	From utility maximization	From $P = MC$ (above AVC)
Curve shape	Downward-sloping 	Upward-sloping 
Movement along	Own price changes	Own price changes
Shift	Income, preferences, other prices	Input costs, technology, # firms
Surplus	Consumer surplus (area below D, above P)	Producer surplus (next lecture!)
Market curve	Horizontal sum of individual D	Horizontal sum of individual S

Summary

Today's Key Takeaways:

1. **The seller's supply rule**: Produce where $P = MC$ on the rising portion of MC, provided $P \geq AVC$
2. This follows directly from the **cost-benefit principle** — produce as long as $P \geq MC$
3. The rule answers three questions: **how much**, **whether**, and **how profitable**
4. **Higher product price** → movement **along** the supply curve (more produced)
5. **Higher input costs** (wages, energy) → MC shifts up → supply curve shifts **left**
6. **Better technology** → MC shifts down → supply curve shifts **right**
7. **Fixed costs don't shift supply** in the short run (they don't affect MC)
8. Supply and demand are **mirror images** — built from the same marginal analysis logic

Connection: We now have the individual firm's supply curve. Next lecture: **producer surplus** and how to build the **market supply** from individual firms.

Next (Lecture 14, April 9): Producer Surplus, Market Supply, and Linear Supply 

Exercises

Practice Time! 

The supply rule, cost changes, and supply shifts.

Exercise 1: Multiple Choice

Question: A sardine restaurant in Lisbon operates in a competitive market. Currently, $P = €12$ per meal and $MC = €10$ at the restaurant's current output level. The restaurant should:

- A. Keep output the same — it is already maximizing profit
- B. Increase output — producing more will add to profit
- C. Decrease output — the restaurant is overproducing
- D. Shut down — it is not covering costs

Answer: B

Since $P = €12 > MC = €10$, each additional meal earns €12 in revenue but costs only €10. The restaurant should **expand output** until MC rises to meet $P = €12$. At the current output, there is still “room” to increase profit by producing more.

Only when $P = MC$ should the firm stop expanding.

Exercise 2: Multiple Choice

Question: The Portuguese government introduces a new tourist tax of €2 per hotel night, paid by the hotel. Which of the following is correct?

- A. The hotel's supply curve shifts right by €2
- B. The hotel's supply curve shifts left (up by €2) — at each price, less is supplied
- C. The hotel moves along its existing supply curve
- D. The hotel's demand curve shifts left

Answer: B

The tax increases the **variable cost** of each room-night by €2. This shifts MC **up** by €2 at every output level, which means the supply curve shifts **left** (equivalently, upward by €2).

At any given market price, the hotel now produces **less** because $P = MC$ is reached at a lower quantity. This is a **shift** of supply, not a movement along it (C is wrong), because the change is in costs, not the product's own price. And it's the supply curve, not demand (D is wrong).

Exercise 3: Open Question

A small boat tour company in Cascais faces the following situation. The market price for a coastal tour is **€40 per ticket**. The firm's cost structure is:

Tours/day (Q)	FC (€)	VC (€)	TC (€)	MC (€/tour)
0	300	0	300	—
1	300	15	315	15
2	300	28	328	13
3	300	39	339	11
4	300	52	352	13
5	300	70	370	18
6	300	96	396	26
7	300	133	433	37
8	300	184	484	51

Hypothetical illustrative example

- Using the $P = MC$ rule, what is the profit-maximizing output at $P = €40$? Calculate profit.
- Now fuel costs rise, adding €10 to the MC of each tour. Write the new MC column. What is the new profit-maximizing output? Calculate the new profit.
- Has the supply curve shifted or has the firm moved along it? Explain.
- At the original cost structure, what is the lowest price at which this firm would still produce? (What is the shutdown price?)
- Draw (describe) what the firm's supply curve looks like: at what price does it “start,” and what does it follow?

Exercise 3: Solution — Part a

a) Apply $P = MC$ on the rising portion:

Q	MC	P vs MC
3	11	$P = 40 > MC \rightarrow$ produce
4	13	$P = 40 > MC \rightarrow$ produce
5	18	$P = 40 > MC \rightarrow$ produce
6	26	$P = 40 > MC \rightarrow$ produce
7	37	$P = 40 > MC \rightarrow$ produce
8	51	$P = 40 < MC \rightarrow$ stop!

$Q^* = 7$ tours/day (last tour where $P \geq MC$).

$$\text{Profit: } \pi = TR - TC = (40 \times 7) - 433 = 280 - 433 = -\text{€}153$$

The firm makes a **loss** of €153/day. But should it still operate? AVC at $Q = 7$: $\frac{133}{7} = \text{€}19$. Since $P = \text{€}40 > AVC = \text{€}19$: **yes, operate!** Shutdown loss would be $-\text{€}300$ (= FC), which is worse.

Exercise 3: Solution — Parts b & c

b) New MC = old MC + €10:

Q	Old MC	New MC	P = 40 vs New MC
1	15	25	40 > 25 → produce
2	13	23	40 > 23 → produce
3	11	21	40 > 21 → produce
4	13	23	40 > 23 → produce
5	18	28	40 > 28 → produce
6	26	36	40 > 36 → produce
7	37	47	40 < 47 → stop!

New $Q^* = 6$ tours/day (down from 7).

New VC at $Q = 6$: old VC + $6 \times €10 = €96 + €60 = €156$. New TC = $300 + 156 = €456$.

New profit: $\pi = (40 \times 6) - 456 = 240 - 456 = -€216$ (loss worsened from $-€153$ to $-€216$).

c) This is a **shift of the supply curve**, not a movement along it. The price didn't change (still €40) — what changed was the **input cost** (fuel). Higher variable costs shift MC upward, which shifts the supply curve **left**. At the same price, the firm now supplies fewer tours (6 instead of 7).

Exercise 3: Solution — Parts d & e

d) The shutdown price = $AVC_{\min}$.

Calculate AVC at each Q:

Q	VC	AVC = VC/Q
1	15	15.00
2	28	14.00
3	39	13.00
4	52	13.00
5	70	14.00
6	96	16.00

Shutdown price = $AVC_{\min}$ = **€13** (at Q = 3–4). Below €13, the firm should produce nothing.

e) The firm's supply curve:

- For $P < €13$: **Q = 0** (shut down)
- For $P \geq €13$: follow the **MC curve on its rising portion** (starting around Q = 3–4)
- As P rises, Q* increases: e.g., P = 18 → Q = 5, P = 26 → Q = 6, P = 37 → Q = 7, etc.

The supply curve **starts at the shutdown point** (P = €13, Q ≈ 3–4) and follows MC upward.

Next Lecture

April 9, 2026: Producer Surplus, Market Supply, and Linear Supply 

We'll introduce **producer surplus** (the supply-side mirror of consumer surplus), build **market supply** from individual firms, and work with **linear supply equations**.

Thank You!

Questions? 🙋

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Next class: Wednesday, April 9, 2026